

UNIVERSITY OF BONN
CAISA LAB

INTRODUCTION TO NATURAL LANGUAGE PROCESSING

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PROJECT REPORT OF TEAM #1

DETECTING MULTILINGUAL & MULTICULTURAL &
MULTI EVENT POLARIZATION OF SOCIAL MEDIA

GROUP MEMBERS:

Doğukan Kıvrak	50334517
Güneş Bade Erdem	50456040
Rafet Filiz	50450457
Buğra Kurgan	50433775

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1. Introduction:

Rapid, unplanned technological advancements, globalization, and increased openness to the world have led people to become quickly attached to technology. As a result, virtual environments have become increasingly preferred for accessing information and news, as well as for socializing. People have begun to prefer expressing themselves on social media platforms rather than face-to-face, for reasons such as ease of access and the anonymity they offer. This situation has given rise to the concept of online polarization, which has become a growing problem manifesting itself in many dimensions, including political ideology, race, religion, gender, and sexual orientation. If the effects of polarization grow faster and more widely than necessary, they can easily undermine a community’s social and cultural values and fuel unrest within the groups that make up that society. This will eliminate the critical intellectual environment that supports social development, leading to society becoming resistant to change. In parallel with these sociotechnical developments, recent advances in artificial intelligence have significantly expanded the capacity of computational systems to model, analyze, and generate human language at scale. Large language models (LLMs) exhibit an increased ability to capture semantic structure, contextual dependencies, and discourse-level patterns [1]. These properties suggest potential utility for analyzing complex linguistic phenomena, such as polarization, which are often implicit, context-dependent, and culturally situated [2]. However, existing evidence also indicates that language models inherit biases from their training data, struggle with domain and distribution shifts, and may conflate legitimate dissent with harmful discourse [3]. In this paper, it is aimed to analyze how effectively the polarization problem can be addressed by training and fine-tuning various carefully selected state-of-the-art language models on the dataset provided for the 9th competition of the SemEval-2026 [4].

2. Related Work:

Automated detection of harmful and polarizing content has been widely studied in the contexts of hate speech, abusive language, and political discourse. Despite strong performance on benchmark datasets reaching up to 87.2% precision in controlled evaluations, these systems generalize poorly to real-world settings, where precision can drop to as low as 9.4% on live social media data [2]. This gap reflects challenges such as label ambiguity, contextual dependence, and domain mismatch.

A key source of error is the difficulty of distinguishing hate speech from offensive but non-hateful language. Studies show that 32.3% of content flagged as hate is actually offensive, indicating that models relying on surface-level cues often conflate profanity with group-targeted harm [2]. Early lexicon-based and n-gram approaches lacked the capacity to capture implicit meaning, sarcasm, or pragmatic context [5]. Neural models such as CNNs and RNNs improved representation learning but still struggled with implicit and context-dependent expressions.

Transformer-based models, including BERT [1] and RoBERTa [6], significantly advanced detection by leveraging contextualized embeddings and long-range dependencies. While effective for explicit hate categories, these models remain challenged by implicit hate estimated at 15.5% of all hate speech, which includes stereotyping, disinformation, conspiracy narratives, and ideological glorification that lack explicit markers [3]. Multitask learning has been proposed to mitigate this issue by jointly modeling related objectives such as hate, offensiveness, stance, and sentiment.

Another major limitation arises from dataset and domain bias. Academic benchmarks overrepresent racial and gender-based hate, whereas real-world polarization is often dominated by political content. Analysis of the HateDay dataset (240,000 tweets across eight languages) reveals strong

cross-cultural variation in both prevalence and targets of hate, as well as substantially lower detection performance for non-European languages [2]. These findings have motivated domain adaptation, continued pretraining, and context-aware modeling that incorporates conversational and structural signals, framing polarization as an emergent property of discourse rather than isolated utterances.

More recently, generative and instruction-tuned language models have been explored for zero-shot and few-shot polarization analysis due to their cross-lingual flexibility [7]. However, concerns remain about calibration, bias amplification, and the ability to reliably distinguish legitimate political critique from harmful polarization. As a result, current research emphasizes structured adaptation, contextual modeling, and rigorous evaluation over relying solely on model scale.

3. Data Exploration and Preprocessing:

The dataset used in this project comes from the POLAR @ SemEval-2026 Task 9 [4], which focuses on detecting multilingual, multicultural, and multievent online polarization. It covers 22 languages and each language contains 3,000–5,000 annotated instances. The dataset includes diverse online platforms such as news websites, Reddit, blogs, Bluesky, and regional forums. The dataset spans multiple sensitive societal contexts, including elections, conflicts, gender rights, migration, religious tensions, and more.

3.1 Data Preprocessing:

To handle the diversity of the POLAR benchmark, datasets from all languages were merged into a single corpus. This allowed models to learn shared multilingual patterns and cross-cultural signals simultaneously. By exposing the models to various linguistic structures and social contexts at once, the goal was to improve generalization across different regions.

3.2 Task Formalization and Annotation:

The dataset is structured into three distinct subtasks to capture different levels of polarization:

- **Subtask 1 (Polarization Detection):** A binary classification task to determine if a text is Polarized (1) or Non-polarized (0).
- **Subtask 2 (Polarization Type Classification):** A multi-label classification task identifying the target of polarization, including Political, Racial/Ethnic, Religious, Gender/Sexual, and Other.
- **Subtask 3 (Manifestation Identification):** A fine-grained multi-label classification of rhetorical strategies, such as Stereotype, Vilification, Dehumanization, Extreme Language, Lack of Empathy, and Invalidation.

3.3 Dataset Statistics and Linguistic Diversity:

Exploratory analysis reveals significant structural and linguistic variation across the 13 primary languages. For example, polarized content is highly imbalanced in languages like Hindi (85.5% polarized) and Hausa (10.7%), while remaining relatively balanced in others like German (47.5%)

and Spanish (50.2%). Average text lengths also vary, from brief 10.8-word snippets in Spanish to more detailed 33.8-character sequences in Chinese.

3.4 Subject Matter and Grammatical Patterns:

The subjects of polarization are highly culturally dependent, with different languages focusing on distinct social and political themes. In German, frequent terms often relate to geopolitics and public health, such as “Russland”, “Krieg”, and “Corona”, whereas English conversations focus heavily on political actors and electoral themes like “Trump”, “Biden”, and “Republicans”. Spanish results demonstrate a shift toward identity, religion, and social issues, prominently featuring words like “judío,” “indígena”, and “feminista”. These thematic differences are mirrored in grammatical patterns: English text contains a high proportion of proper nouns due to the frequent mention of political figures, while German utilizes a denser concentration of nouns and verbs, suggesting a more descriptive and content-heavy style. Such variations underscore the critical need for models that can adapt to unique cultural and linguistic structures rather than relying on a universal approach.

4. Methodology:

In this project, multilingual transformer-based NLP techniques were used for the POLAR benchmark analysis, focusing on polarization detection, type classification and manifestation identification. Feature extraction was performed using DistilMBERT, mBERT, mDeBERTa, RemBERT, and XLM-RoBERTa. Models were optimized using the Binary Cross-Entropy with Logistic loss function and the AdamW optimizer. Performance was evaluated using Binary F1 and Macro F1 scores to ensure robust assessment across binary and multi-label classification tasks.

4.1 Task Design and SemEval:

The methodology addresses the hierarchical complexity of the SemEval 2026 POLAR benchmark through three distinct levels of analysis. Subtask 1 is formulated as a binary classification problem to detect the presence of polarization in multilingual social media text. Subtasks 2 and 3 extend this into a multi-label classification framework, identifying specific polarization types such as political or religious. Their rhetorical manifestations including dehumanization and out-group bias. This multi-label design is essential to capture instances where a single document exhibits overlapping polarization characteristics.

4.2 Used Models:

The selection of model architectures was strategically designed to evaluate a spectrum of multilingual capabilities, ranging from distilled baselines to state-of-the-art transformers. DistilMBERT[8] was initially utilized, a lightweight model that leverages knowledge distillation to retain approximately 92% of mBERT’s performance while being 40% smaller and 60% faster. This served as the primary efficiency benchmark[9]. This was compared with the foundational mBERT, which consists of 12 layers and 110 million parameters trained on 104 languages. Despite being the standard baseline for multilingual NLP, mBERT’s shared vocabulary space can occasionally lead to

“language interference” requiring the inclusion of more advanced architectures in the evaluation suite.

To explore the impact of architectural refinements on polarization detection, mDeBERTa was integrated. Unlike standard BERT-based models, mDeBERTa employs a disentangled attention mechanism, which represents each word using two separate vectors that one for its content and one for its relative position[10]. This allows the model to better capture the structural differences in how polarization is expressed across different languages. Furthermore, XLM-RoBERTa was employed, which improves upon the BERT architecture by removing the Next Sentence Prediction (NSP) objective and training on a significantly larger, 2.5TB filtered CommonCrawl dataset[11]. The robustness of XLM-RoBERTa’s pre-training makes it particularly effective at handling the diverse and often informal linguistic styles found in social media discourse. The most sophisticated model in the pipeline is RemBERT, a 579-million-parameter architecture specifically optimized for cross-lingual transfer[9]. RemBERT differentiates itself through a significantly expanded vocabulary and a reconfigured embedding structure where the output embeddings are larger than the input embeddings. This design choice allows the model to learn more refined semantic representations during pre-training, which is critical for the POLAR benchmark’s multi-label subtasks. By utilizing such a high-capacity model, the aim was to address the label ambiguity and cultural nuances inherent in the dataset, as RemBERT’s depth allows it to capture more abstract sociopolitical cues that smaller models might overlook.

Task	Lang.	Monolingual						Cosslingual					
		InfoXLM	LaBSE	RemBERT	XLM-R	mBERT	mDeBERTa	InfoXLM	LaBSE	RemBERT	XLM-R	mBERT	mDeBERTa
1. Polarization	Amharic	77.96	78.92	81.93	80.64	53.71	74.98	23.26	49.44	5.63	0.00	0.00	83.21
	Arabic	58.39	61.94	67.19	70.42	52.48	54.68	29.68	35.87	27.76	2.78	37.79	25.63
	English	72.09	69.84	75.43	76.08	72.77	74.94	1.90	51.68	53.41	46.89	33.44	53.45
	German	29.75	63.46	67.50	67.78	58.45	64.38	0.62	51.97	64.59	41.87	46.05	25.94
	Hausa	59.57	66.41	67.43	66.41	59.32	65.62	8.21	22.92	6.47	1.50	6.25	25.13
	Spanish	38.69	64.04	70.98	57.00	59.31	54.16	45.70	68.66	67.26	21.09	68.34	62.17
	Urdu	1.07	66.35	79.95	43.91	65.63	60.68	1.60	30.36	0.00	0.00	3.68	68.67
2. Types	Amharic	24.22	38.13	43.65	25.71	17.93	25.00	11.56	20.65	2.49	0.00	0.00	26.76
	Arabic	22.97	40.23	42.12	37.58	27.53	35.22	11.49	23.73	8.78	2.11	12.47	7.93
	English	16.04	23.25	31.38	24.10	21.07	17.70	10.84	19.69	16.42	12.22	11.70	3.93
	German	13.52	58.58	61.19	58.56	54.13	40.64	12.03	35.86	29.59	9.72	23.98	11.88
	Hausa	18.56	19.14	17.38	18.09	19.61	18.87	3.20	9.97	3.91	1.39	5.78	5.90
	Spanish	43.26	66.07	67.76	58.07	57.94	43.40	7.89	47.06	15.38	0.43	32.02	14.60
	Urdu	27.14	51.94	51.60	45.38	38.02	33.13	3.77	13.62	6.33	0.00	3.94	20.30
3. Manifestations	Amharic	43.52	47.56	47.63	43.17	33.18	43.29	15.57	27.07	8.64	0.00	0.00	43.58
	Arabic	40.05	51.55	52.52	55.61	42.18	47.73	16.56	30.68	22.14	0.00	19.57	17.24
	English	14.40	15.01	19.39	18.61	18.60	15.15	7.16	10.16	10.05	5.62	10.69	8.85
	German	38.49	49.88	52.74	51.70	46.85	51.91	2.38	36.12	27.05	0.00	23.38	12.93
	Hausa	19.23	20.04	19.18	18.89	18.74	18.93	5.74	5.86	3.77	3.12	6.19	6.43
	Spanish	38.94	50.00	51.04	45.02	45.17	35.09	2.34	40.63	11.83	0.40	35.99	23.56
	Urdu	34.26	52.20	53.64	41.32	45.90	47.01	2.10	19.16	11.54	0.00	1.88	48.58

Figure 1: Table Of Results From POLAR [9]

4.3 Development Environments and Programming Languages:

All components were developed using the Python programming language, chosen for its robust ecosystem in machine learning and data science. The project primarily utilized the PyTorch framework for model construction and training, interfaced through the Hugging Face Transformers and Datasets libraries. This setup enabled efficient model loading, tokenization and fine-tuning across various architectures. Data manipulation and result visualization were handled using NumPy, Pandas and Matplotlib, providing a streamlined pipeline from initial data preprocessing to final

performance analysis within a GPU-accelerated computing environment.

5. Experimental Setup:

This section describes the experimental setup used to train and evaluate the multilingual transformer models across the three POLAR subtasks. All experiments were conducted using the official datasets released for SemEval-2026 Task 9 (POLAR)[4], with the goal of ensuring a fair and controlled comparison between models. The dataset splits differ across subtasks according to the benchmark specification. For Subtask 2, the data were divided into training, validation, and test sets following an approximate 70/15/15 split, while the datasets for Subtask 1 and Subtask 3 were divided into two parts using a 70/30 split. The subtasks also differ in formulation and complexity: Subtask 1 is framed as a binary classification task distinguishing polarized from non-polarized content, whereas Subtasks 2 and 3 are formulated as multi-label classification tasks in which each instance may be associated with multiple polarization targets or manifestations.

Five multilingual transformer-based models were evaluated: DistilMBERT, mBERT, mDeBERTa, XLM-RoBERTa, and RemBERT. These models were selected to cover a range of architectural sizes and multilingual pretraining strategies, enabling analysis of how model capacity and pretraining scale influence polarization detection performance. All models were fine-tuned end-to-end using a task-specific classification head, with a single sigmoid output for Subtask 1 and independent sigmoid outputs for each label in Subtasks 2 and 3. To ensure comparability across models, a uniform training configuration was applied, using the AdamW optimizer with a learning rate of $2e-5$, a batch size of 32, and a maximum sequence length of 128 tokens. Binary cross-entropy with logits was used as the loss function for all tasks.

Subtasks 1 and 2 were trained for five epochs, as validation performance typically stabilized early, while Subtask 3 was trained for up to ten epochs due to the higher complexity of manifestation detection. Model checkpoints were selected based on validation performance, using the highest validation Binary F1-score for Subtask 1 and the highest validation Macro F1-score for Subtasks 2 and 3. For inference, a default probability threshold of 0.5 was used, and label-specific threshold tuning was applied on the validation set for Subtask 2 and selected labels in Subtask 3 to improve recall for low-frequency categories. All experiments were implemented in Python using the Hugging Face Transformers library, and training logs were recorded at the epoch level to support reproducibility and detailed analysis.

6. Evaluation and Results:

This section presents the experimental results for the three POLAR subtasks. All models were evaluated under identical conditions, with performance reported on held-out test sets to ensure a fair comparison.

Different scores were used depending on the subtask. For Subtask 1, performance is measured using the Binary F1-score to balance precision and recall for detecting polarized content. For Subtasks 2 and 3, the Macro F1-score serves as the primary metric, as it assigns equal weight to each label and is suitable for handling class imbalance.

6.1 Performance Highlights Across Subtasks:

Subtask 1 yielded the strongest and most stable results across all architectures. Even lightweight models like DistilMBERT achieved competitive scores (0.797 Binary F1), indicating that coarse-grained polarization signals are relatively easy to capture. Notably, mDeBERTa slightly outperformed larger models like RemBERT in this binary classification with 0.823 Binary F1-score.

In Subtask 2, categorizing the target of polarization (e.g., Political, Religious, Racial) proved more difficult due to label overlap and class imbalance. While Political polarization was the easiest to identify, the “Other” category consistently performed poorly, staying below 0.50 F1 for all models. RemBERT achieved the highest overall score (0.659 F1), though its advantage over smaller models remained modest.

Subtask 3 represents the most significant challenge in the benchmark, as it requires detecting implicit rhetorical strategies rather than simple keywords. Models are generally more successful at identifying explicit linguistic markers, such as vilification (0.712 F1) or stereotypes (0.664 F1). However, they struggle significantly with subtle pragmatic cues, such as dehumanization (0.520 F1) or a lack of empathy (0.540 F1), which often rely on context rather than overt vocabulary. Among the evaluated architectures, RemBERT achieves the strongest performance with overall 0.618 F1-score, though the incremental gains suggest that no model has fully mastered these complex manifestations.

6.2 Summary of Results:

In short, performance dropped as the tasks became more detailed. While larger models like RemBERT generally lead the way, they don’t solve the core issues of label ambiguity and class imbalance. This suggests that the primary challenge appear to be dataset-level factors, such as label ambiguity and cross-lingual variability, rather than model capacity alone.

7. Analysis and Discussion:

By examining performance shifts across the three subtasks, the effects of task formulation, data properties, and model capacity on multilingual polarization detection were explored. A clear performance gradient appears across the subtasks. Binary polarization detection in Subtask 1 consistently yields strong results across all models, showing the presence of explicit linguistic cues and allowing quick convergence with minimal overfitting. In contrast, polarization type classification in Subtask 2 and manifestation identification in Subtask 3 show progressively lower Macro F1 scores, revealing increased semantic and pragmatic complexity. In addition, the multi-label setting and significant class imbalance mainly limited the performance in Subtask 2. The Political polarization is the easiest category to detect, where the other label performed significantly worse due to a poorly organized definition. These issues become more evident in Subtask 3, where models must identify implicit and context-based rhetorical strategies. Manifestations such as dehumanization and lack of empathy remain challenging across all architectures, highlighting constraints in capturing pragmatic and discourse-level signals.

A key strength of this project is the controlled comparison of multiple multilingual transformer models under the same conditions, enabling a fair evaluation of architectural differences. However, gains from larger models are modest in detailed tasks, suggesting that dataset challenges outweigh model capacity. Despite these limitations, the results show that multilingual transformers work

well for broad polarization detection, but fine-grained predictions should be treated cautiously in field use.

8. Conclusion:

In this project, a systematic evaluation of multilingual transformer-based models on the POLAR benchmark is conducted to investigate online polarization across three increasingly fine-grained subtasks. The results show that binary polarization detection is a stable and well-solved task under the current benchmark. All evaluated models show strong, consistent performance on Subtask 1, with quick convergence and minimal overfitting, indicating that polarization is often signaled by explicit linguistic cues. In contrast, polarization type classification and manifestation identification remain substantially more challenging. Subtask 2 is primarily constrained by class imbalance and label ambiguity, particularly in the other category, whereas Subtask 3 highlights the difficulty of detecting implicit, context-dependent rhetorical strategies, such as dehumanization and a lack of empathy. Overall, the research confirms that although multilingual transformer models provide a strong basis for polarization analysis, performance on fine-grained tasks is mainly limited by dataset-level factors rather than by the model architecture alone. Thus, the study objective of evaluating the strengths and drawbacks of current multilingual models on the POLAR benchmark was successfully achieved.

9. Future Work:

The results of this project point to several directions to improve fine-grained multilingual polarization analysis. A primary focus should be on data quality and label design. Refining annotation guidelines, breaking down distinct categories like ‘other’, or introducing structured label structures could reduce ambiguity and strengthen learning stability, especially for multi-label subtasks. From a modeling perspective, future efforts might explore label-aware and context-sensitive approaches, such as modeling label correlations, using multi-task learning with related objectives, such as stance or sentiment, or incorporating discourse-level representations. These techniques can more precisely capture implicit and overlapping polarization patterns that are difficult to model with standard classification heads. Additional improvements could come from cross-lingual adaptation and domain-specific pretraining, especially for under-resourced languages and culturally diverse contexts. Finally, expanding evaluation beyond isolated text classification to include conversational or temporal scenarios, along with more detailed qualitative error analysis and explainability techniques, could provide a greater understanding of how polarization manifests and evolves in real-life online discussions.

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